

Reinforcement Learning and PK-PD Models Integration to Personalize the Adaptive Dosing Protocol of Erdafitinib in Patients With Metastatic Urothelial Carcinoma

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The integration of pharmacokinetic-pharmacodynamic (PK-PD) modeling and simulations with artificial intelligence/machine learning algorithms is one of the most attractive areas of the pharmacometric research. These hybrid techniques are currently under investigation to perform several tasks, among which precision dosing. In this scenario, this paper presents and evaluates a new framework embedding PK-PD models into a reinforcement learning (RL) algorithm, Q-learning (QL), to personalize pharmacological treatment. Each patient is represented with a set of PK-PD parameters and has a personal QL agent which optimizes the individual treatment. In the training phase, leveraging PK-PD simulations, the QL agent assesses different actions, defined consistently with the clinical knowledge to consider only plausible dose-adjustments, in order to find the optimal rules. The proposed framework was evaluated to optimize the erdafitinib treatment in patients with metastatic urothelial carcinoma. This drug was approved by the US Food and Drug Administration (FDA) with a dose-adaptive protocol based on monitoring the levels of serum phosphate, which represent a biomarker of both treatment efficacy and toxicity. To evaluate the flexibility of the methodology, a heterogeneous virtual population of 141 patients was generated using an erdafitinib population PK (PopPK)-PD literature model. For each patient, treatment response was simulated by using both QL-optimized protocol and the clinical one. QL agents outperform the approved dose-adaptive rules, increasing more than 10% the efficacy and the safety of treatment at each end point. Results confirm the great potentialities of the integration of PopPK-PD models and RL algorithms to optimize precision dosing tasks.

Study Highlights

WHAT IS THE CURRENT KNOWLEDGE ON THE TOPIC?

✓ The integration of pharmacometric models with reinforcement learning (RL) algorithms for treatment personalization in realistic case studies is still limited.

WHAT QUESTION DID THIS STUDY ADDRESS?

✓ By combining modeling and simulation with artificial intelligence (AI) techniques, is it possible to achieve a better pharmacological response with respect to clinical practice? What are the advantages in using an RL-based individualized dosing strategy rather than the same set of rules for the whole population? To answer these questions in a realistic case study, erdafitinib treatment in patients with metastatic urothelial cancer was considered. The proposed methodology was directly compared with erdafitinib dose-adaptive protocol approved by the US Food and Drug Administration (FDA).

WHAT DOES THIS STUDY ADD TO OUR KNOWLEDGE?

✓ This AI/modeling and simulation technique improved the response to erdafitinib treatment by efficiently tackling patient interindividual variability. Even if further extensions and evaluations are necessary to refine the methodology here presented and to test it on different case studies, this work can represent a guideline to start from.

HOW MIGHT THIS CHANGE CLINICAL PHARMACOLOGY OR TRANSLATIONAL SCIENCE?

✓ Pharmacometric models can inform precision dosing. By integrating RL algorithms in this workflow, it is possible to optimize individual treatment reaching more efficiently targeted end points.

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The role of artificial intelligence (AI) and machine learning (ML) in the clinical pharmacology field is constantly increasing. AI/ML are now present throughout the entire continuum of pharmacology research, from the early drug discovery phases^{1–3} to the clinical practise.^{4,5} Recently, hybrid approaches integrating AI/ML with mathematical modeling and simulations have been proposed and gained great momentum in the pharmacometric landscape to solve different tasks^{6–10} such as pharmacokinetic (PK) and pharmacodynamic (PD) predictions,^{6,7} covariate selection^{6,7,10,11} as well as treatment optimization, especially in a precision dosing context.¹²

Precision dosing refers to tailoring the dose on each individual patient at each given time, in order to ensure the greatest benefits and least risks. It is recommended for classes of compounds characterized by a narrow therapeutic index and/or high interindividual variability (IIV).^{13–18} In these cases, an adaptive dosing protocol is typically adopted where patient health status is periodically monitored through measurements of efficacy/toxicity biomarkers,^{14–16} and dose consequently adjusted to maintain these biomarkers within target ranges, similarly to the therapeutic drug monitoring practise.¹⁹

Reinforcement learning (RL)²⁰ is a branch of ML which naturally fits to the problem of precision dosing based on periodically patient monitoring.^{21–24} Indeed, RL techniques train a computer (i.e., agent) to select the best action to perform on a system based on its current state in order to steer the system to a target condition.^{21,22,25} The learning process leverages a trial-and-error interaction between system and agent: at each system state, the agent tries different actions on the system and evaluates their consequences (i.e., new system state) based on a reward signal quantifying the suitability of the performed action to achieve the final goal. In the context of precision dosing, the agent represents the clinician that periodically monitors the patient status through efficacy/safety biomarkers (i.e., system state) and consequently selects the dose to administer (i.e., action). Patient response in terms of biomarker level (i.e., new system state) is observed and the suitability of the administered dose (i.e., selected action) is quantified based on distance from the target (i.e., collection of the reward).

Training an RL algorithm to solve a precision dosing task ideally requires a high amount of retrospective data in treated patients that must include a multitude of dosing scenarios. This could limit the actual applicability of the approach in the real world. To address this issue, a number of recent works has promoted the idea of coupling RL algorithms with population PK-PD modeling.^{21,23,25,26} In this context, PK-PD models were used to simulate virtual patients and, thus, to generate experience from different dosing scenarios on which the RL algorithm could learn the optimal dosing strategy. In almost all the works reported in the literature, a unique RL-agent was trained to find an optimal dosing strategy for an entire patient population. Most of the applications fall inside the oncology field^{15,18–20} where, for instance, this approach was used to optimize neutrophil counts²⁷ or to reduce the tumor size in patients undergoing chemotherapy.^{28–30} Despite the promising results, the number of works applied to real clinical scenario is still limited.²⁴

In this paper, a new framework combining RL and PK-PD modeling was proposed to improve precision dosing of erdafitinib (Balversa, Janssen Pharmaceutical companies), an anticancer

compound approved by the US Food and Drug Administration (FDA) for the treatment of metastatic urothelial carcinoma^{31–34} Erdafitinib is characterized by a narrow therapeutic window and high IIV of the response.³² The serum phosphate concentration,³⁵ $[\text{PO}_4]_{\text{serum}}$, is the most important efficacy and toxicity biomarker.^{31,32,36} In particular, $[\text{PO}_4]_{\text{serum}} \geq 5.5$ mg/dL correlated with a better response to treatment,³¹ whereas $[\text{PO}_4]_{\text{serum}} < 7$ mg/dL represents the safety threshold for hyperphosphatemia, the major cause of drug-related adverse events.³¹ Consequently, erdafitinib administration requires an adaptive-dosing strategy aiming to maintain the $[\text{PO}_4]_{\text{serum}}$ level within its target range (i.e., 5.5, 7 mg/dL). Despite the good performances of the approved adaptive dosing protocol, the narrow therapeutic windows and the significant IIV of pharmacological response suggested that the erdafitinib therapy could potentially benefit from a personalization of the dose adjustment rules at the individual patient level. Therefore, in this work, the proposed RL-approach was challenged to individually optimize the adaptive dosing protocol of erdafitinib by tailoring the dose adjustment rules on each patient. To this end, the presented methodology leverages the concept of digital twins,^{37,38} a virtual replica of the patient here represented by the PK-PD model describing the $[\text{PO}_4]_{\text{serum}}$ dynamics^{32,39} with its set of individual covariates and parameters. For each virtual patient, a personal RL-agent was trained to individually optimize the erdafitinib protocol by maximizing efficacy and reducing the risk of adverse events. The performances of the RL-based individual protocols were compared with the FDA approved dose-adjustment rules which provided the benchmark for the evaluation.

METHODS

Reinforcement learning

RL encloses different algorithms for solving sequential decision-making problems.²⁰ The mathematical framework of RL is represented by a finite Markov Decision Process (MDP) which formalizes the sequential decision-making scenario.^{40,41}

As illustrated in **Figure 1**, an MDP is characterized by a decision maker, called agent, that interacts with a system or environment to achieve a given goal.

In a finite MDP, the system is described by a set of N states, $S = \{s_1, \dots, s_N\}$, and the agent by a set of M possible actions, $A = \{a_1, \dots, a_M\}$, to perform. It is assumed that the agent-system interaction occurs at each step of a finite time sequence, $t = 0, 1, 2, \dots, T$. At each step t , the agent observes the system state, $S_t = s$, and, consequently, performs an action, $A_t = a$, on the system. The assignment of an action a to a given state s is formally represented by a map, named policy, $\pi(s) = a$. At step $t + 1$, the system evolves to the next state, S_{t+1} , and the agent receives a reward, R_{t+1} , in response to its previous action. The reward is feedback in terms of a real number, which scores the goodness of the agent action. R_{t+1} is given by a reward function, potentially depending on S_t, S_{t+1}, A_t . Reward function is a crucial part of an MDP formalization and needs to be carefully designed.

Rewards are returned by the system to the agent after every action, resulting in a sequence, $R_{t+1}, R_{t+2}, \dots, R_T$, whose values are higher if the performed actions are useful to reach the target. From a mathematical perspective, in an MDP, the agent has to maximize the discounted return, G_t (Eq. 1), which depends on all the rewards collected in all subsequent time steps.

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^T \gamma^k R_{t+k+1} \quad (1)$$

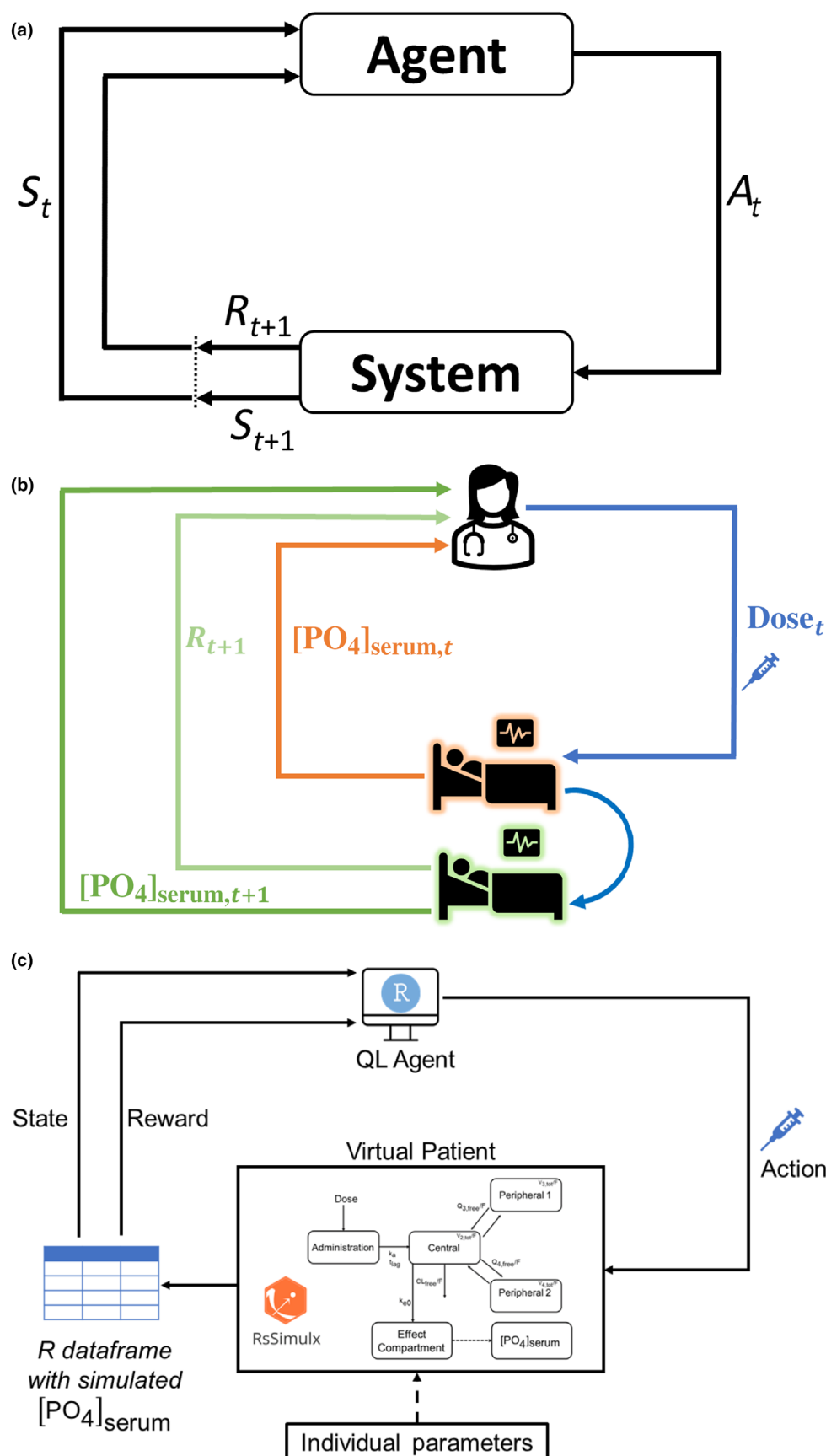


Figure 1 Mapping of a MDP (a) to precision dosing based on patient monitoring (b). This clinical MDP can be implemented within the RL framework as illustrated in (c). MDP, Markov Decision Process; RL, reinforcement learning. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/cpt.3176)]

As reported in Eq. 1, G_t depends on the discounted rate, $\gamma \in (0, 1)$, a parameter that ensures the convergence of G_t for $T \rightarrow \infty$ and regulates the importance given to future rewards with respect to the present one. If γ is close to 0, the agent will maximize only the immediate reward, on the contrary, if $\gamma \rightarrow 1$, the agent becomes more farsighted.

Solving an MDP implies finding the optimal policy, $\pi^*(s) = a^*$, which allows the agent to take the best action, a^* , for each state $s \in S$ in order to maximize G_t . To this aim, different RL algorithms were developed.^{20,23} Most of them relies on a utility function, called Q-function or action-value function, $Q^\pi(s, a)$, to estimate the optimal policy. As described in Eq. 2, $Q^\pi(s, a)$ is defined as the conditional expected value of G_t given the choice of action a when system state is s , according to the policy π .

$$Q^\pi(s, a) := \mathbb{E}_\pi \left[G_t | S_t = s, A_t = a \right] = \mathbb{E}_\pi \left[\sum_{k=0}^T \gamma^k R_{t+k+1} | S_t = s, A_t = a \right]. \quad (2)$$

Following Eq. 2, the optimal policy π^* maps each state s with the best action $a^* = \arg \max_a Q^\pi(s, a)$.

Q-Learning (QL) is an RL algorithm²⁰ which provides a discrete approximation in a tabular form of the Q-function for each state-action couple. Given a set of N states and M actions, QL returns a matrix, $\mathbf{Q} \in \mathbb{R}^{N \times M}$, in which the element $Q(s_i, a_j)$ represents an approximation of the Q-function for action a_j in state s_i . As illustrated in Figure S1.1, a set of decision-making rules can be easily derived from the $\mathbf{Q}$ matrix.

Erdafitinib case study

Erdafitinib is an orally administered inhibitor of the fibroblast growth factor receptors (FGFR) approved for the treatment of metastatic urothelial carcinoma.^{31–34} $[\text{PO}_4]_{\text{serum}}$ level is the most important efficacy/safety biomarker for erdafitinib treatment.^{31,32,36} Indeed, $[\text{PO}_4]_{\text{serum}}$ increases as a consequence of the FGFR target engagement^{33,42,43} and a $[\text{PO}_4]_{\text{serum}}$ level $> 5.5 \text{ mg/dL}$ was found to correlate with a better response in terms of progression-free survival.³¹ At the same time, hyperphosphatemia, i.e., an excessive $[\text{PO}_4]_{\text{serum}}$ level ($\geq 7 \text{ mg/dL}$) is the major cause of drug-related adverse events.³¹ Consequently, the FDA-approved erdafitinib protocol is based on an adaptive dosing strategy (Figure 2) aiming to maintain the $[\text{PO}_4]_{\text{serum}}$ within the target range (5.5, 7 mg/dL). Briefly, 5 discrete dose levels (i.e., 4, 5, 6, 8 and 9 mg/day), are available. Treatment always starts at a dose of 8 mg/day. After 2 weeks (the 14th day) of treatment, patients are monitored and the initial dose is increased to 9 mg/day in case of insufficient efficacy (i.e., $[\text{PO}_4]_{\text{serum}} < 5.5 \text{ mg/dL}$). Patient monitoring typically occurs on a monthly basis. In case of hyperphosphatemia events, erdafitinib

administration is temporarily interrupted, patient monitoring becomes weekly, and treatment is resumed when $[\text{PO}_4]_{\text{serum}}$ returns $< 5.5 \text{ mg/dL}$ (Figure 2a). The resumption dose depends on the hyperphosphatemia severity, discretized in 3 grades, that is, 7, 9 mg/dL, 9, 10 mg/dL, and $> 10 \text{ mg/dL}$, and on the duration of the treatment interruption (Figure 2b).

A population PK-PD model describing the erdafitinib PK and its effect on $[\text{PO}_4]_{\text{serum}}$ is available.^{32,39} Model equations and population parameter values are reported in Supplementary Material S2. Based on this PK-PD model, an *in silico* evaluation of the adaptive dosing protocol approved for erdafitinib was performed.³² It revealed that, after 4 months of treatment (i.e., the median time for assessing treatment efficacy³²), the 30% of patients was expected to have $[\text{PO}_4]_{\text{serum}}$ within the target range, 5.5, 7 mg/dL. It also highlighted a high IIV of pharmacological response.³²

RL-based precision dosing of erdafitinib

The erdafitinib precision dosing problem was modeled through an MDP (Figure 1c). It requires an accurate definition of three key elements, that is, system states, agent actions, and reward function, that have to be designed coherently with the clinical application. System state, that is, patient status, is described by $[\text{PO}_4]_{\text{serum}}$ which is periodically monitored to guide dose adjustment (i.e., agent action). As the clinical goal is keeping $[\text{PO}_4]_{\text{serum}}$ within the target range of 5.5, 7 mg/dL, thus guaranteeing a phosphate level above the efficacy threshold and simultaneously avoiding hyperphosphatemia events, dose adjustments are evaluated by a reward function depending on the $[\text{PO}_4]_{\text{serum}}$ level. The RL-agent was trained through a QL algorithm, by using a digital patient twin, represented by the erdafitinib PK-PD model, that links drug doses with $[\text{PO}_4]_{\text{serum}}$ level, with its set of individual parameters and covariates.

Reward function. The reward function was a $\mathbb{R}^D \rightarrow \mathbb{R}$ function of the $\mathbb{R}^D$ vector, $[\mathbf{PO}_4]_{\text{obs}} = ([\text{PO}_4]_{\text{obs},1}, \dots, [\text{PO}_4]_{\text{obs},D})$, containing D daily $[\text{PO}_4]_{\text{serum}}$ observations, where D depends on the number of days between 2 consecutive patient monitoring actions, that is, 7, 14, or 28 days. It is given by the sum of two contributes weighted by the coefficients β_1 and β_2 (Eq. 3):

$$\text{Reward}([\mathbf{PO}_4]_{\text{obs}}) = \beta_1 \cdot \sum_{i=1}^D g([\text{PO}_4]_{\text{obs},i}) + \beta_2 \cdot \frac{\sum_{i=1}^D h([\text{PO}_4]_{\text{obs},i})}{D} \quad (3)$$

where the expressions of the functions g and h are reported in Eqs. 4 and 5, respectively.

$$g([\text{PO}_4]_{\text{obs},i}) = \begin{cases} 0.5 \cdot \exp\left(-\lambda_2 \cdot \left[-([\text{PO}_4]_{\text{obs},i} - 5.5)\right]\right) & \text{if } [\text{PO}_4]_{\text{obs},i} < 5.5 \text{ mg/dL} \\ \min\left(1, \left([\text{PO}_4]_{\text{obs},i} - 5.5\right) + 0.5\right) & \text{if } [\text{PO}_4]_{\text{obs},i} \in [5.5, 6.25] \text{ mg/dL} \\ \min\left(1, \left([\text{PO}_4]_{\text{obs},i} - 6.5\right) + 1\right) & \text{if } [\text{PO}_4]_{\text{obs},i} \in (6.25, 7) \text{ mg/dL} \\ 0.5 \cdot \exp\left(-\lambda_1 \cdot \left([\text{PO}_4]_{\text{obs},i} - 7\right)\right) & \text{if } [\text{PO}_4]_{\text{obs},i} \in [7, 9] \text{ mg/dL} \\ 0.18 \cdot \exp\left(-\lambda_2 \cdot \left([\text{PO}_4]_{\text{obs},i} - 9\right)\right) & \text{if } [\text{PO}_4]_{\text{obs},i} > 9 \text{ mg/dL} \\ \lambda_1 = 0.5, \quad \lambda_2 = 1.5 & \end{cases} \quad (4)$$

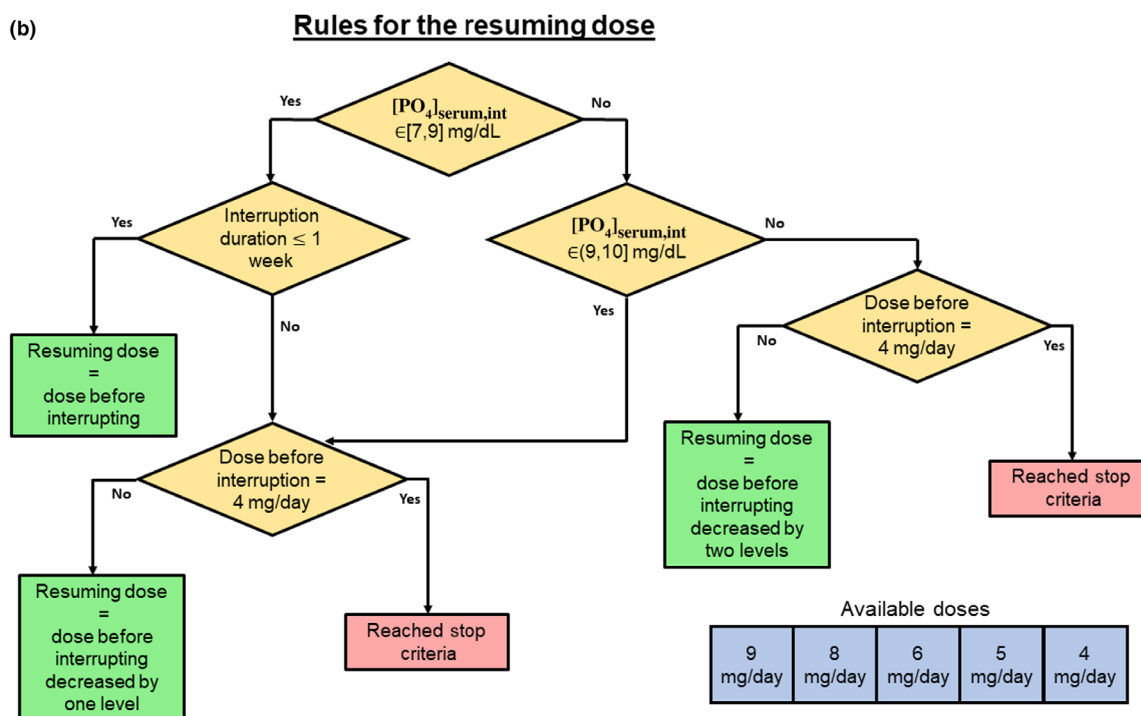
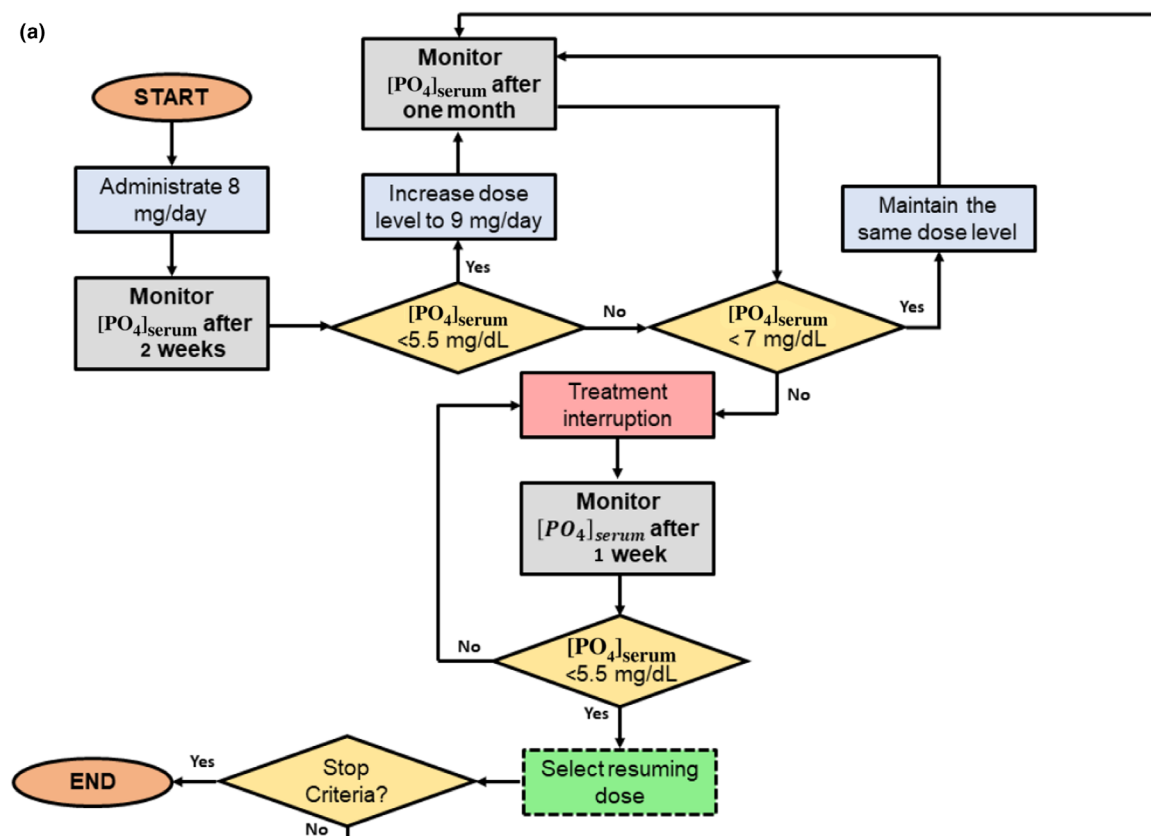


Figure 2 Flow chart of the FDA approved adaptive-dosing protocol of erdafitinib. Dose levels are adjusted according to the $[PO_4]_{\text{serum}}$ level (a). In case of treatment interruption, the resuming dose is selected based on the severity of the hyperphosphatemia event and on the interruption duration (b). FDA, US Food and Drug Administration. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/cpt.3176)]

$$b([PO_4]_{obs,i}) = \begin{cases} 1 & [PO_4]_{obs,i} \in [5.5, 7) \text{ mg/dL} \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

The first term in Eq. 3 accounts for the $[PO_4]_{serum}$ levels as transformed by function g . As illustrated in Figure 3, it assigns higher reward to actions bringing $[PO_4]_{serum}$ close to the efficacy range with the maximum remuneration for the 6.25, 7 mg/dL interval because a higher $[PO_4]_{serum}$ is associated with a better response in terms of overall survival and progression-free survival.³¹ Conversely, actions leading to hyperphosphatemia or inefficacy are penalized. The penalty becomes more significant as the biomarker moves further away from the target window. The second term assigns an extra reward proportional to the percentage of days in a treatment cycle in which $[PO_4]_{serum}$ is maintained in the 5.5, 7 mg/dL range. The coefficients β_1 and β_2 were arbitrary fixed to 10 and 5, respectively. In this way, the first term in Eq. 1, that represents the core of the reward function, has a higher weight than the second one, that was introduced to stress the importance of maintaining $[PO_4]_{serum}$ in the target range and to increase the gap, in terms of reward, between actions leading to efficacy and those bringing $[PO_4]_{serum}$ to toxicity or inefficacy.

System states. The state of the system (patient) is described by a tuple of three elements, $X = \{L_{PO_4}, D_p, L_{int}\}$. The first, L_{PO_4} , accounts for the $[PO_4]_{serum}$ observation at the monitored day, discretized in 6 levels, as detailed in Eq. 6:

$$L_{PO_4}([PO_4]_{serum}) = \begin{cases} 1 & \text{if } [PO_4]_{serum} < 5.5 \text{ mg/dL (inefficacy)} \\ 2 & \text{if } [PO_4]_{serum} \in [5.5, 6) \text{ mg/dL (efficacy-1)} \\ 3 & \text{if } [PO_4]_{serum} \in [6, 6.5) \text{ mg/dL (efficacy-2)} \\ 4 & \text{if } [PO_4]_{serum} \in [6.5, 7) \text{ mg/dL (efficacy-3)} \\ 5 & \text{if } [PO_4]_{serum} \in [7, 9) \text{ mg/dL (moderate hyperphosphatemia)} \\ 6 & \text{if } [PO_4]_{serum} > 9 \text{ mg/dL (severe hyperphosphatemia)} \end{cases} \quad (6)$$

In particular, the efficacy range 5.5, 7 mg/dL was subdivided into 3 intervals to allow the QL agents to select different actions depending on whether the $[PO_4]_{serum}$ is located in the lower ([5.5, 6) mg/dL), the middle ([6, 6.5) mg/dL), or the upper ([6.5, 7) mg/dL) part of the target window. The second element of patient state, D_p , codes for the last administered dose. Only the 5 clinically available dose levels, that is, 4, 5, 6, 8, and 9 mg/day, were considered. Dose level is coupled with a sign (+/-) that is negative if the treatment is temporary interrupted, and positive otherwise (Eq. 7). This was an arbitrary convention adopted to store two information, that is, the temporary interruption and the triggering dose, in a compact form. Indeed, storing information on the dose leading to an interruption is necessary to choose the resumption dose. A flag value equal to -1 encodes for the treatment beginning in which the initial dose must be determined.

$$D_p = \begin{cases} -1 & \text{for the initial state} \\ -4, -5, -6, -8, -9 & \text{for doses provoking a temporary treatment interruption} \\ 4, 5, 6, 8, 9 & \text{otherwise} \end{cases} \quad (7)$$

Finally, L_{int} stores the discretized $[PO_4]_{serum}$ level (Eq. 8) that caused a temporary treatment interruption, $[PO_4]_{serum, int}$. As the severity of hyperphosphatemia triggering treatment interruption is a key clinical factor for the choice of restarting the treatment and of the resuming dose, L_{int} was defined to distinguish between severe ($L_{int} = 6$) and moderate ($L_{int} = 5$) conditions. A flag value -1 was arbitrary assigned to states not associated to a temporary interruption.

$$L_{int}([PO_4]_{serum}) = \begin{cases} 5 & \text{if } [PO_4]_{serum, int} \in [7, 9) \text{ mg/dL} \\ 6 & \text{if } [PO_4]_{serum, int} > 9 \text{ mg/dL} \\ -1 & \text{no interruption} \end{cases} \quad (8)$$

An initial set of 60 couples $\{L_{PO_4}, D_p\}$ was defined by combining all the L_{PO_4} values with each $D_p \neq -1$. Couples associated with treatment interruption, i.e., with $D_p < -1$, were combined with all the positive values of L_{int} ; conversely, the ones with $D_p > 0$ were combined only with $L_{int} = -1$, obtaining 90 different states. Because the temporary treatment interruption causes a $[PO_4]_{serum}$ decrease, interruption states with a post-discontinuation $[PO_4]_{serum}$ greater than the before-discontinuation $[PO_4]_{serum}$ (i.e., $L_{PO_4} > L_{int}$) were not considered. Thus, the tuples with $D_p < -1$, $L_{PO_4} = 6$ and $L_{int} = 5$ were removed, reducing the state number to 85. Because at the beginning of

the treatment all patients have $[PO_4]_{serum} < 5.5$ mg/dL,³² only one initial state with $L_{PO_4} = 1$, i.e. $X_I = \{1, -1, -1\}$, was added to the states space, thus leading the final number of states to 86.

Agent actions. Actions were designed to give the agent more freedom of choice with respect to the clinical protocol rules in order to allow the definition of potentially better personalized treatments. However, safety constraints derived from the clinical practice, such as gradual dose variation, treatment interruption due to toxicity, or dosage reduction post-interruption, were taken into account and summarizes all the possible agent actions stratified by system states. Differently from the clinical protocol (Figure 2), the agent can select the initial dose among all the possible dose levels and decide to increase ($D+$)/decrease ($D-$) by one level or maintain ($D=$) the previous administered dose at each patient

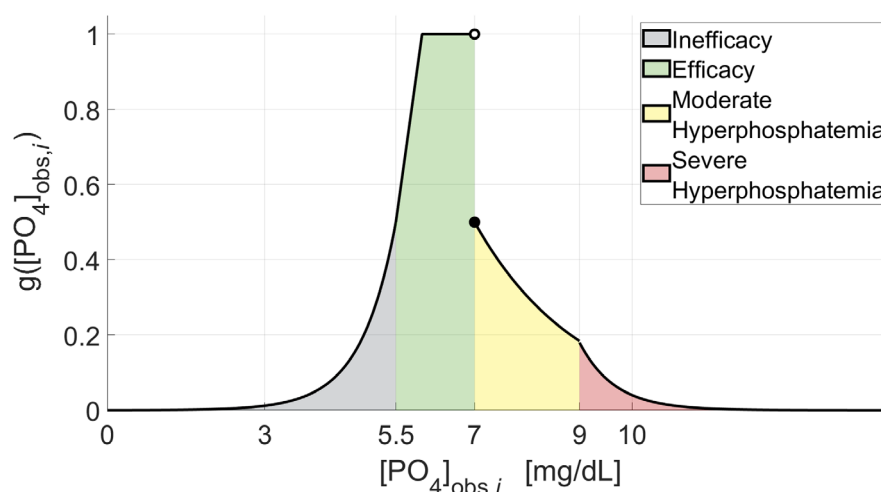


Figure 3 Reward assigned to daily observed $[PO_4]_{\text{serum},i}$, $[PO_4]_{\text{obs},i}$ with function g (Eq. 4). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/cpt.3176)]

monitoring step, in absence of hyperphosphatemia. Adverse events are differently tackled by the agent according to their severity. In case of moderate hyperphosphatemia ($L_{PO_4} = 3$), the agent can decide to temporarily interrupt the treatment (Int) or to continue erdafitinib administration with a dose lower or equal to the previous one, D_p . In case of severe hyperphosphatemia ($L_{PO_4} = 4$), the agent is constrained to interrupt the treatment. After that, treatment can be resumed with the dosage-rules reported in Table 1 only if $L_{PO_4} < L_{\text{int}}$, otherwise interruption is maintained. Conversely, clinical protocol enables treatment resumption only if $[PO_4]_{\text{serum}} < 5.5 \text{ mg/dL}$ ($L_{PO_4} = 1$).

Implementation. As illustrated in Figure 1c, a simulation platform that iteratively predicts the response to adaptive dosing administration of erdafitinib was embedded within the QL-framework to allow the continuous interaction between the QL-agent and the virtual patient. The algorithm was coded in R version 3.4.1 (Comprehensive R Network, <http://cran.r-project.org/>). The simulation platform was developed based on the erdafitinib PK-PD model^{32,39} (Supplementary Material S2), coded in Mlxtran and simulated through the R package RsSimulx (Simulx 2021R1, Lixoft SAS, a Simulations Plus company). At each treatment cycle, the QL-agent decided the dose level for the next cycle on the basis of the patient status, the $[PO_4]_{\text{serum}}$ response was simulated by the model and the simulated output used to update the patient status and compute the reward.

The same workflow was used for simulating the FDA-approved protocol. In this case, the decisional engine is the adaptive dosing rules reported in Figure 2.

All simulations were performed without considering residual unexplained variability (RUV).

Evaluation of the methodology. The proposed QL-setup was challenged to personalize a 5-month treatment (i.e., average duration of therapy⁴⁴) in a virtual population of 141 patients generated by the stratified random sampling strategy described in Supplementary Material S3. Individual parameters and covariates of virtual patients were extracted from erdafitinib population PK/PD model and the covariates distribution of subjects on which the model was estimated (see Supplementary Materials S2 and S3). The stratified sampling was applied to obtain a heterogeneous virtual population in terms of treatment response (see Supplementary Material S3) on which the proposed methodology was evaluated. As detailed in Supplementary Material S3, $[PO_4]_{\text{serum}}$ level

can be within the target range at treatment end only for completely responsive subjects. Conversely, erdafitinib becomes no longer effective before the fifth month for partially responsive individuals. The virtual population includes both completely ($n = 96$) and partially ($n = 45$) responsive subjects. An individual QL-agent was trained for each virtual

Table 1 Summary of the agent actions stratified by each system/patient state

D_p	Allowed actions
Initial state	
-1	4, 5, 6, 8, 9 [mg/day]
States without hyperphosphatemia: $L_{PO_4} \leq 4$ and $L_{\text{int}} = -1$	
4	$D+$, $D =$
5	$D+$, $D-$, $D =$
6	$D+$, $D-$, $D =$
8	$D+$, $D-$, $D =$
9	$D-$, $D =$
States with moderate hyperphosphatemia: $L_{PO_4} = 5$ and $L_{\text{int}} = -1$	
4	Int., 4 [mg/day]
5	Int., 4, 5 [mg/day]
6	Int., 4, 5, 6 [mg/day]
8	Int., $D = 4, 5, 6, 8$ [mg/day]
9	Int., 4, 5, 6, 8, 9 [mg/day]
States with severe hyperphosphatemia: $L_{PO_4} = 6$ and $L_{\text{int}} = -1$	
For each D_p	Int.
States in which treatment can be resumed: $L_{PO_4} < L_{\text{int}}$ and $L_{\text{int}} > 0$	
-4	0, 4 [mg/day]
-5	0, 4, 5 [mg/day]
-6	0, 4, 5, 6 [mg/day]
-8	0, 4, 5, 6, 8 [mg/day]
-9	0, 4, 5, 6, 8, 9 [mg/day]
States in which treatment cannot be resumed: $L_{PO_4} = L_{\text{int}}$ and $L_{\text{int}} > 0$	
For each D_p	0 mg/day

Table 2 Comparison between the clinical and QL-based individual protocols at timepoints of interest

Range of $[\text{PO}_4]_{\text{serum}}$ [mg/dL]	%Patients in the whole population [141 subjects]		%Patients in the subgroup of completely responsive patients [96 subjects]	
	Clinical protocol	QL-based individual protocols	Clinical protocol	QL-based individual protocols
Evaluation at the second week of treatment				
<5.5	1.42	2.13	2.08	3.13
[5.5, 7)	34.05	76.59	31.25	77.08
[7, 9]	48.93	21.28	50.00	19.79
>9	15.60	0.00	16.67	0.00
Evaluation at the end of the fourth month of treatment				
<5.5	17.73	0.00	6.25	0.00
[5.5, 7)	70.21	97.87	76.05	96.87
[7, 9]	12.06	2.13	17.70	3.13
>9	0.00	0.00	0.00	0.00
Evaluation at the end of treatment (fifth month)				
<5.5	36.88	31.91	7.29	0.00
[5.5, 7)	56.73	68.09	83.33	100
[7, 9]	6.39	0.00	9.38	0.00
>9	0.00	0.00	0.00	0.00

Percentages of patients with $[\text{PO}_4]_{\text{serum}}$ within a certain range were used as evaluation metric. Results are reported for both the overall population and the subgroup of responsive patients.

patient to optimize erdafitinib adaptive dosing protocol at the individual level using the hyperparameters reported in [Table S1.1](#). The individual QL-based protocols were compared with the FDA-approved protocol ([Figure 2](#)) and with a population QL-based protocol learned by training a single QL-agent for the entire population (i.e., one-for-all QL agent) of patients (see [Supplementary Material S5](#)).

RESULTS

For each subject of the virtual population ($n = 141$), a QL-agent was trained to personalize the erdafitinib adaptive dosing protocol on a time-window of 5 months. To obtain a valid evaluation benchmark for the QL-based individual protocols, a 5-month treatment was simulated on the same virtual population following the FDA-approved adaptive dosing rules. As reported in [Table 2](#), the 2 strategies were compared in terms of percentage of patients with $[\text{PO}_4]_{\text{serum}}$ in the ranges of inefficacy (< 5.5 mg/dL), efficacy (5.5, 7 mg/dL), moderate (7, 9 mg/dL), and severe (> 9 mg/dL) hyperphosphatemia. Three clinically relevant timepoints were considered: the second week of treatment, that is, first patient evaluation, the end of the fourth month, that is, median time for assessing erdafitinib efficacy,³² and the end of treatment.

At each of the three timepoints, the QL-based individual protocols resulted in a higher percentage of patients with $[\text{PO}_4]_{\text{serum}}$ in the efficacy range and in a lower percentage of subjects in hyperphosphatemia than the clinical one. In particular, at the end of the fourth month, following the QL-based protocols 98% of patients was in the efficacy window (vs. 70.21% for the clinical protocol). Prolonging the treatment, erdafitinib lost effectiveness for some patients in the partially responsive group, therefore a drop in the percentage of subjects having $[\text{PO}_4]_{\text{serum}} \in [5.5, 7)$ mg/dL was observed for both the strategies (68.09% vs. 56.73% for QL-based

and clinical protocol, respectively). However, in the responsive patient subgroup, this decrease was not observed, confirming the out-performances of the QL-agents in maintaining the $[\text{PO}_4]_{\text{serum}}$ in target.

In addition, efficacy and safety of the individual QL-based protocols and the clinical one were compared on the entire treatment period. In particular, paired analyses of the percentages of days in which individuals have $[\text{PO}_4]_{\text{serum}} < 5.5$ mg/dL, $[\text{PO}_4]_{\text{serum}} \in [5.5, 7)$ mg/dL and $[\text{PO}_4]_{\text{serum}} \geq 7$ mg/dL were performed at patient level applying a Wilcoxon paired signed rank test ($\alpha = 0.05$, H_0 = median of the differences between percentages in the two groups is equal to 0). As illustrated in [Figure 4a–c](#), H_0 was always rejected. Therefore, on a treatment window of 5 months, QL-agents were able to maintain $[\text{PO}_4]_{\text{serum}}$ in the target range for longer time, minimizing its permanence in both inefficiency and toxicity ranges. For what regards toxicity, the QL-based protocols completely avoided the onset of severe toxicity events and drastically reduced the number of moderate hyperphosphatemia occasions ([Figure 4d](#)). An example is reported in [Figure 5a](#).

The personalization of the initial dose significantly increased QL-agent performances. As reported in [Figure 4e](#), the clinical initial dose of 8 mg/day was selected by the QL-agents only in 23.40% of patients, whereas the lower starting doses were generally preferred. [Figure 5b](#) reported the example of a completely responsive patient with a 9 mg/day ideal dose ([Supplementary Material S3](#)): its optimized treatment started with a 4 mg/day dose that was then gradually increased up to 9 mg/day. This strategy allowed to circumvent the hyperphosphatemia events that occurred following the clinical protocol. A gradual dose increase was mainly adopted

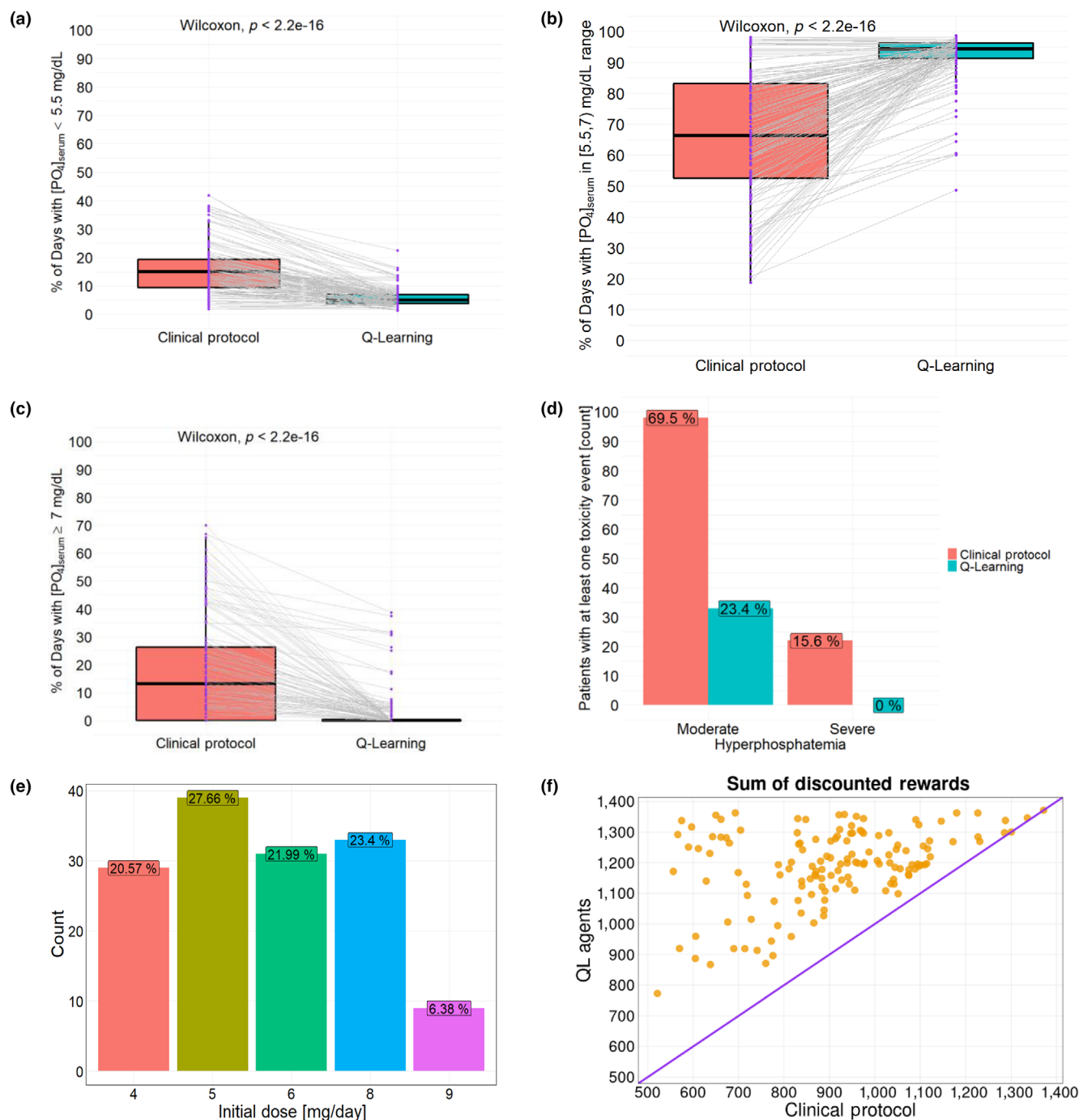


Figure 4 Summary of the performances of QL-based individual protocols. (a–c) Paired comparisons of QL-agents and the clinical protocol on the entire treatment time period for the percentages of days in which patients have $[PO_4]_{\text{serum}} < 5.5 \text{ mg/dL}$ (a), $[PO_4]_{\text{serum}} \in [5.5, 7) \text{ mg/dL}$ (b) and $[PO_4]_{\text{serum}} \geq 7 \text{ mg/dL}$ (c). For each of them, a Wilcoxon paired signed rank test ($\alpha = 0.05$) was performed. (d) Paired comparisons of QL-agents and the clinical protocol for the number of moderate and severe hyperphosphatemia events. (e) Distribution of the starting doses selected by the QL-agents (8mg/day is the clinical starting dose). (f) Paired comparison of QL-agents and the clinical protocol in terms of reward function. QL, Q-learning. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/cpt.3176)]

by QL-agents also for partially responsive patients as it kept $[PO_4]_{\text{serum}}$ in target for a longer time, until erdafitinib lost its effectiveness (Figure S4.1). These examples also underlined the strength of allowing QL-agents to change the dose at each monitoring step.

Finally, under the assumption that the reward function (Eqs. 3–5) is a suitable measurement of the achievement of the therapeutic goal of erdafitinib therapy, the sum of discounted rewards obtained with the QL-based individual and clinical protocols were compared. As reported in Figure 4f, the QL-agents

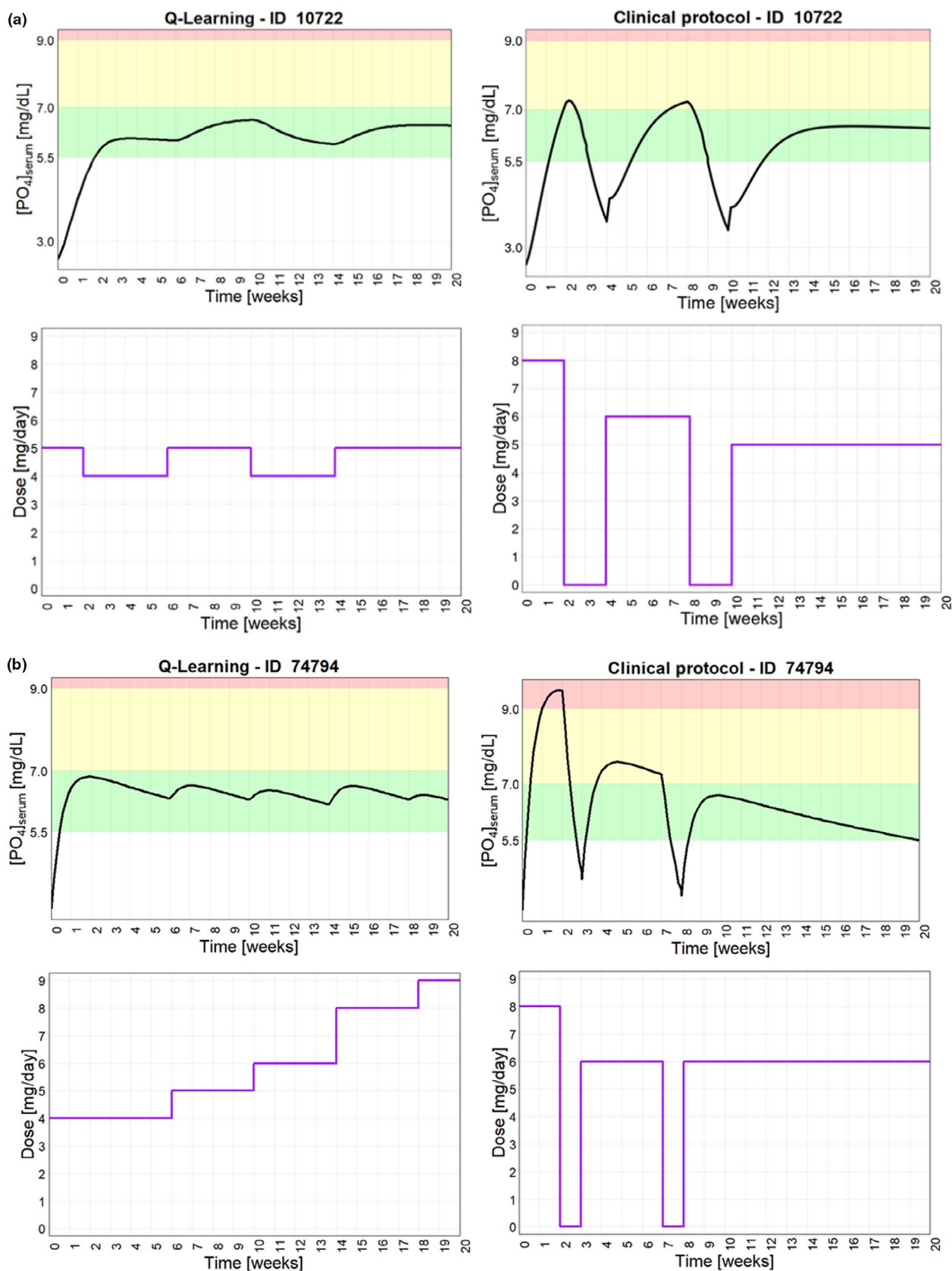


Figure 5 (a) Main results of dose individualization strategies based on QL agents. Panel (a) and (b) provide two examples of how QL agent can improve erdafitinib treatment as compared with FDA approved dose adjustment protocol. FDA, US Food and Drug Administration; QL, Q-learning. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/cpt.3176)]

performed better, or at least equal, than the FDA approved protocol.

DISCUSSION

A recent trend in pharmacometrics is coupling PK-PD models with AI/ML methods to perform several tasks.^{6–9} In particular, the integration of RL and PK-PD modeling is currently under investigation for precision dosing purposes and has already demonstrated interesting performances in preliminary studies, especially within the oncologic field.^{24,27–30} However, applications of this methodology to real clinical precision dosing problems are still limited.²⁴

In this work, a new precision dosing framework coupling PK-PD models and an RL algorithm, the Q-learning, was developed and applied on a relevant case study directly derived from clinical oncology, that is, the erdafitinib-based treatment of urothelial carcinoma. An adaptive dosing protocol (Figure 2) in which erdafitinib dose is adjusted based on $[PO_4]_{\text{serum}}$, the efficacy/safety biomarker of this treatment, is already available. However, due to the narrow therapeutic window and significant IIV of pharmacological response,³² erdafitinib therapy could potentially benefit from a personalization of the adaptive dosing protocol at individual patient level. The proposed RL-approach was developed with this aim. Indeed, differently from the previous works^{26,28,30} where the objective was the identification of an optimal dosing strategy for an entire patient population, here, the scope of the RL-approach was to individually optimize the erdafitinib adaptive dosing protocol by tailoring the dose adjustment rules on each patient. In the RL-context, this change of perspective means to move from a unique RL-agent, trained on an entire patient population, to a set of personal QL-agents, each trained on a single individual patient.

The successful application of an RL-based approach to the individualization of an adaptive dosing protocol relies on some fundamental steps that are here discussed.

First, the clinical precision dosing problem must be correctly formalized in terms of elements composing the RL-framework, that for QL are system states, agent actions, and reward function. More in detail, system states must include all the information relevant to take the right action to perform (dose selection in this study). For the erdafitinib case study, states included the discretized $[PO_4]_{\text{serum}}$ level representing the patient status but also stored information on the previous administered dose and the hyperphosphatemia level that caused temporary treatment interruption. In particular, $[PO_4]_{\text{serum}}$ was discretized in six levels, one for inefficacy, three for efficacy, and two for moderate and severe toxicity. Splitting the efficacy $[PO_4]_{\text{serum}}$ range in multiple sublevels allowed the QL-agent to select a different action depending on whether the biomarker lies in the bottom, middle, or top of the target window. This was fundamental to adequately manage the loss of erdafitinib efficacy over time. Regarding the agent actions, they must account for relevant clinical constraints, for example, due to safety reasons, in order to obtain QL-based protocols acceptable in the clinical practice. At the same time, agents must have a certain degree of freedom in making decisions to allow a treatment customization on individual patients.

For erdafitinib accordingly to the clinical protocol, QL-agents were imposed to select only clinically available doses, to gradually increase/decrease the dose and to temporally interrupt the treatment in case of severe hyperphosphatemia. Instead, differently from the clinical protocol, QL-agents were allowed to select a personalized initial dose potentially different from 8 mg/day, to gradually adjust erdafitinib dose at each monitoring occasion, to decrease the dose without interrupting the treatment in the presence of moderate toxicities, and to resume treatment after interruption if the hyperphosphatemia severity is decreased at least by one level. The set of actions that the agent can perform was designed with the aim of obtaining clinically acceptable protocols. However, their choice was arbitrary and it could be of interest to train QL-agents with a different degree of freedom. Finally, the reward function must suitably encode the therapeutic scope of the treatment to correctly score the different actions and optimize the treatment. For erdafitinib, the goal was to maintain $[PO_4]_{\text{serum}}$ within its efficacy range (i.e., 5.5, 7 mg/dL) as long as possible, avoiding hyperphosphatemia events. The reward function was designed to faithfully summarize the expert knowledge on erdafitinib treatment available from the literature.^{31,32,35,39,44} Its form was selected based on previous experience on other case studies⁴⁵ and it does not simply describe the difference from a target value.^{30,46,47} As the adopted reward function led to satisfactory results, a fine-tuning process was not performed. However, in view of an actual clinical application, the design of the reward function, as well as the entire formalization of the proposed RL-framework, should be carefully reviewed with clinicians to ensure that all the key aspects of the erdafitinib treatment are correctly considered and modeled. In addition, the reward could be further expanded, for example, including terms linked to the probability of survival or of adverse reactions, leading to a multi-objective treatment optimization.

Second, the QL-framework must be integrated with a simulation engine to *in silico* train the QL-agents without the need of a high amount of retrospective clinical data. Following the approach proposed in ref. [26], in this work, the simulation engine was based on a population PK-PD model^{32,39} which allowed to simulate patient responses to all the possible dosing scenarios, even those particularly hazardous for the patient that could not be tested in the clinical setting,²⁶ thereby providing a more robust training of the QL-agents. In addition, as the aim was to individually optimize the administration protocol for each individual patient, the PK-PD model combined with a set of patient-specific individual parameters represented the digital patient twin needed to train the patient-specific QL-agent.

Once the RL-based precision dosing framework has been developed, it must be tested on a heterogeneous (virtual) population, in terms of pharmacological response. Indeed, evaluating the RL performances on very different scenarios is fundamental to assess its flexibility with respect to the IIV, a central feature in precision dosing which goal is finding the right dosage rules for every patient. In this work, a heterogeneous virtual population ($n = 141$) was *ad hoc* constructed equally sampling patients from both completely ($n = 96$) and partially ($n = 45$) responsive groups further stratified by their ideal dose levels. This strategy allowed to design a virtual

population including a wide range of different patients, maintaining a limited sample size.

The individual QL-based protocols outperformed the FDA-approved one in terms of both efficacy and safety. They resulted able to lead $[PO_4]_{\text{serum}}$ in the target range in a higher patient percentage and to maintain the biomarker in target for a longer time-period respect to the clinical protocol. In addition, the severe hyperphosphatemia events were completely avoided, and the moderate ones drastically reduced. The personalization of the starting dose was fundamental for the nice performances of the QL-agents that were able to detect patients tolerating the highest dose (Figure S4.4) as well as those requiring a low initial drug amount (Figures S4.1 and S4.3, Figure 5) or a loading dose (Figure S4.2). These findings suggested that for erdafitinib the individualization of the starting dose could significantly contribute to reduce toxicity events, especially at the beginning of the treatment. Further, also a gradual dose variation at each monitoring step, even when the biomarker falls in the target window, was a further key feature of QL-agents that allowed a better and more far-sighted control of $[PO_4]_{\text{serum}}$. This was a deviation from the clinical strategy which is based on pushing high doses and then lowering them if toxicity events occur. Differently, in almost all the occasions of moderate hyperphosphatemia, QL-agents confirmed protocol decision of interrupting treatment rather than continuing with a lower or equal drug level (Figure S4.5).

Overall, the obtained finding demonstrated that the erdafitinib treatment could significantly benefit from a precision dosing strategy in which dose adjustment rules are individually tailored on each individual patient. This conclusion was further confirmed by comparing the individual QL-based protocols here identified by training a personal QL-agent for each patient with an optimal QL-based adaptive protocol obtained by training a single QL-agent on the entire patient population (one-for-all QL-agent), see **Supplementary Material S5**. The protocol learned by the one-for-all QL-agent performed worse than the individual-based QL-agents. Indeed, the output of the one-for-all QL-agent is a general dosing strategy that is not individually tailored on each subject but provides individualized dosing indications learned by evaluating their effect on a group of patients rather than on a single one. Therefore, it can work well for the majority of the population but cannot be the optimal solution for some of the patients. This limitation is imputable to the RL approach itself, but it is intrinsic to every dosing strategy designed for a whole population.

Despite the RL-based dosing suggestions could contribute to generate new advancements in the personalization of erdafitinib treatment, the proposed methodology relies on some relevant assumptions that partially limit the immediate clinical applicability of the approach as it is and that have to be accurately addressed before the presented findings could successfully translate into an actual clinical protocol. First of all, it was hypothesized that the PK-PD model correctly described the erdafitinib pharmacological response and RUV was not considered in the simulations. Absence of RUV is a strong simplification that was necessary for a preliminary comprehensible evaluation of the methodology here proposed. Indeed, as it is known that effectiveness of RL algorithms

potentially decreases when RUV is introduced,²⁶ RUV must be adequately managed, with solutions that must depend on what RUV represents (e.g., measurement errors or model misspecification). In the first case, moving from MDP to Partially Observable MDP⁴⁸ could be a possible strategy to formally describe the uncertainty of system state, due to RUV. However, model misspecification should be addressed differently. In the second case, the reward function would be no longer deterministic, and it could be necessary to switch from classical RL algorithms (e.g., QL) to a different version that properly handles this stochastic framework.^{20,48} Furthermore, it was assumed that digital patient twins were well-characterized since the beginning of the treatment (i.e., the individual PK/PD model parameters were known). However, individual parameter estimation requires the availability of individual data and, consequently, needs to be performed during the monitoring of the treatment. The integration of the QL-based treatment optimization within a Bayesian paradigm that allows continuous learning of model parameters from the monitoring observations,^{45,49,50} could definitively cope with this issue. In this scenario, at each monitoring step, the individual model used as the specific patient digital twin and the personal RL-agent should be updated accounting for new patient data. This means that the personal RL-agent has to be continuously re-trained accordingly on the new available patient knowledge. As this process makes a massive usage of model simulations, it can be very time-consuming in the presence of a complex modeling framework. The duration of the retraining process, depending on the clinical workflow, has to fall within the time frame spanning from patient observation to clinical decision in order to leverage RL-based suggestion. Consequently, an efficient implementation of the algorithm and the modeling framework on a powerful computing architecture are needed.

In conclusion, coupling PK-PD with RL algorithms can provide a useful approach to support precision dosing and improve treatment personalization. However, a few more steps forward are necessary to completely exploit the potentiality of these techniques in real clinical pharmacology scenarios.

SUPPORTING INFORMATION

Supplementary information accompanies this paper on the *Clinical Pharmacology & Therapeutics* website (www.cpt-journal.com).

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CONFLICT OF INTEREST

All other authors declared no competing interests for this work.

AUTHOR CONTRIBUTIONS

A.D.C., E.M.T., and P.M. wrote the manuscript. A.D.C., E.M.T., M.F., and P.M. designed the research. A.D.C. and M.F. performed the research. A.D.C., E.M.T., and M.F. analyzed the data.

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